A reduced-order thermal model and multi-input controller for the HISPEC cryostat

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Cryogenic astronomical instruments such as the HISPEC spectrograph must hold large optical assemblies at a stable operating temperature near 55 K, yet their thermal behavior is set by hundreds of thousands of coupled conduction and radiation pathways, far too many states to simulate in real time or to design a controller against. This project develops an end-to-end pipeline that turns a CAD assembly into a lumped thermal-network model and reduces it to a compact form suitable for on-board control. From the instrument geometry an octree mesh builds a multi-million-node thermal graph; the linearized dynamics are projected onto their 120 slowest physical modes and then reduced by balanced truncation to a 30-state model that keeps only the jointly controllable and observable dynamics. On this reduced model I design a multi-input linear-quadratic-regulator controller with a static state estimator and a regularized steady-state feedforward for its 33 heaters and 28 controlled sensors. The 30-state model reproduces the full plant's heater-to-sensor step response to within about 0.15%, small enough to design against while remaining light enough to run on a microcontroller. Remaining work adds surface-to-surface radiation and hardware deployment.